

Kannataki HW-2

$$1) \mathcal{T}_B' = \{U \subseteq X : \forall x \in U, \exists B_x \in \mathcal{B} \ni x \in B_x \subseteq U\}$$
$$\mathcal{T}_B = \left\{ \bigcup_{C \in \mathcal{C}} C : \mathcal{C} \subseteq \mathcal{B} \right\}$$

Claim: They are equivalent

Pf: Claim (i) $\mathcal{T}_B' \subseteq \mathcal{T}_B$

Proof: Let $U \in \mathcal{T}_B'$. For each $x \in U$, we know $\exists B_x \in \mathcal{B}$ s.t. $x \in B_x \subseteq U$. Thus, $\bigcup_{x \in U} B_x = U$. Hence, as each $B_x \in \mathcal{B}$, we have $U \in \mathcal{T}_B$ as well, thus, $\mathcal{T}_B' \subseteq \mathcal{T}_B$ as claimed.

Claim (ii): $\mathcal{T}_B \subseteq \mathcal{T}_B'$

Proof: Let's fix a family $\mathcal{C} \subseteq \mathcal{B}$, & consider $F = \bigcup_{C \in \mathcal{C}} C$. Let $x \in F$, then, $\exists C_x \in \mathcal{C}$ s.t. $x \in C_x$. As C_x is a basic element, we know that $C_x \subseteq F$, hence $F \in \mathcal{T}_B'$!

Thus, $\mathcal{T}_B \subseteq \mathcal{T}_B'$.

Combining, we have $\mathcal{T}_B = \mathcal{T}_B'$.

2) (i) $\mathcal{T}_C \subseteq \mathcal{T}_X$

Pf: Let U be open, i.e., $U = \bigcup_{C \in \mathcal{C}} C$, each C be open &

satisfying the gluing property. As arbitrary union of open sets is open, hence U is open in X , i.e., $\mathcal{T}_C \subseteq \mathcal{T}_X$.

(ii) $\mathcal{T}_X \subseteq \mathcal{T}_C$

Pf: Let U be open in X . Then for any $x \in U$, $\exists C_x \in \mathcal{C}$ s.t. $x \in C_x \subseteq U$.

Subclaim: $\bigcup_{x \in U} C_x = U$ | Pf: $\exists$. Initially, $\bigcup_{x \in U} C_x \subseteq U$ as $C_x \subseteq U \forall x \in U$

Also, as $x \in C_x \subseteq U$, thus $\bigcup_{x \in U} C_x \supseteq U$, thus, we have $\bigcup_{x \in U} C_x = U$, & thus, $\mathcal{T}_X \subseteq \mathcal{T}_C$. We are done!!

T.S.:

3) $\mathcal{B} = \{B_A : A \subseteq \mathbb{Z}\}$ is a basis of X

Let $f \in X$ be a function. Let $Z(f)$ be the set of its zeroes. Then, to verify:

- (i) for any function $f \in X$, $\exists$ a basic element $B_f \ni f$.
 (ii) for $B_{A_1} \cap B_{A_2} \neq \emptyset$, if $g \in B_{A_1} \cap B_{A_2}$, then $\exists B_{A_3}$ s.t.
 $g \in B_{A_3} \subseteq B_{A_1} \cap B_{A_2}$

Pf: (i) Obvious, just consider $B_{Z(f)} \in \mathcal{B}$.

(ii) Again obvious, set $A_3 = A_1 \cup A_2$, & hence $g \in B_{A_3} \subseteq B_{A_1} \cap B_{A_2}$

4) (a) Again, verify using basis properties.

(i) Pick any $m \in \mathbb{Z}$. Then, as $m = m \cdot 1 + 0$, hence $m \in B_{m,0}$, a basic element.

(ii) If $B_{d_1, r_1} \cap B_{d_2, r_2} \neq \emptyset$, say, $m \in B_{d_1, r_1} \cap B_{d_2, r_2}$,

that means $\exists n_1, n_2$ s.t. $d_1 n_1 + r_1 = d_2 n_2 + r_2 = m$.

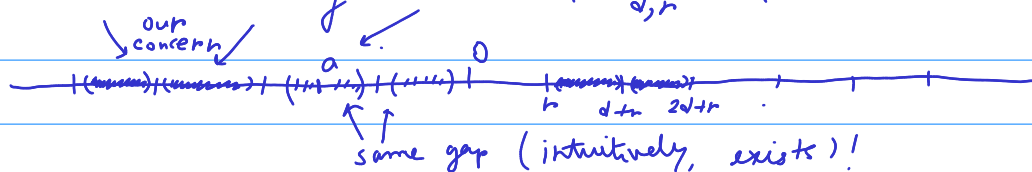
consider $B_{\gcd(d_1, d_2), m}$, it works for the given property!

(b) Suppose $FTS \cup \subset \exists \mathcal{U} = \{a_1, a_2, \dots, a_n\}$ which is open.

Then $\exists B \in \mathcal{B}$ s.t. $a_1 \in B \subseteq \mathcal{U}$. However, B is infinite while $\mathcal{U}$ is finite $\rightarrow$ contradiction!

(c) $\mathbb{Z}/B_{d,r}$ open $\rightarrow$ i.e., to show that $B_{d,r}$ is simultaneously open & closed, as $B_{d,r}$ is a basic element!

Pick any $a \in \mathbb{Z}/B_{d,r}$.



In fact, this is exactly the union of the residue classes (mod d) except $r \pmod{d}$! $\rightarrow$ i.e. $\mathbb{Z}/B_{d,r} = \bigcup_{\substack{i \in \mathbb{Z} \\ 0 \leq i \leq d-1 \\ i \neq r}} B_{d,i}$, and as arbitrary union of basic elements, hence open!!

Hence, this is totally disconnected!

(d) Lastly, this is also Hausdorff! $\rightarrow$

Claim: For $a, b \in \mathbb{Z}$, $\exists$ open U_a & U_b s.t. $a \in U_a, b \in U_b$,
& $U_a \cap U_b = \emptyset$.

V

Pf:



Eg: As we showed that $\mathbb{Z} \setminus B_{d,n}$ is also open,

If we make sure $b \notin U_b^c \subseteq U_a$ we are done!

(& this is easy! choose $U_a = B_{d',n}$ & U_b^c
which doesn't include b , & take its complement

$$\left(\begin{array}{c} a \\ \cdot \end{array} \right) \quad \cdot \quad b$$

I.S. $\rightarrow$

(5) $S_p = \{n \in \mathbb{Z}^+ : p|n\}$. Then,

$$S = \{S_p : p \text{ is prime}\} \cup \{1\}$$

is a sub-basis on $\mathbb{Z}^+$.

Eg Check: If not, then $\exists n \in \mathbb{Z}$, thus n is a
prime, hence $n \in S_n \in S$, ($\Rightarrow \Leftarrow$)

(6 & 7) $\rightarrow$ Bland Exercises.